

SCI-JINC

Scientific Rationale and Contract

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Status. Implementation-blind Stage B author draft. The scientific substance has not yet completed manager review, scientific-owner review, implementation-blind consistency review, or freeze. No implementation conformity, representation fidelity, validation, achieved performance, numerical readiness, production readiness, or production claim is made.

Input	Exact admitted PTC transformed occurrences and positive JINC-permitted PTC coefficients; exact same-processed-sample AST coordinates; one authorized array-associated JINC parameter set; target WCS; finite positive processed-sample frequency.
Output	Zero through three independent observation bundles, at most one for each requested and admitted stable array. Each complete bundle contains N_p , C_p , Q_p , derived m_p with local support/validity, and $T_p^{(\kappa^2)}$.
Core equation	$m_p = (\sum_i I_{ip} \omega_i \kappa_{ip} z_i) / (\sum_i I_{ip} \omega_i \kappa_{ip})$.
Scientific sources	Approved SCI-JINC Scope Brief and sanitized boundary packet; recovered implementation-independent signed-estimator core under its supersession cover; Schloerb LMT OTF/JINC method excerpt under its TolTEC-exclusion cover.
Availability	The generic scientific contract is defined. No TolTEC numerical parameter set and no registered, selected, realized JINC-permitted PTC coefficient family/payload is supplied by the author packet, so no numerical route is claimed by this draft.

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1 Program adherence and prior-work recovery

This package follows the Citlali Scientific Contract Library Program, its pilot process controls, and the owner-approved downstream sequence recorded by the sanitized Scope Brief. It begins after the frozen SCI-MAP scientific authority without altering it. The author received only the exact approved SCI-JINC packet and the exact external objects admitted through packet covers. No implementation, configuration, test, audit, repair, validation, reduction, status, production, or manager-side scientific material was consulted.

Prior work is reused rather than repeated. The recovered signed estimator with distinct numerator, signed normalization, and quadratic accumulator is adopted. The generic OTF/JINC analytic family is cited through the admitted Schloerb method excerpt [3]. The earlier circular radial-cutoff and pixel-area-average branches are superseded by the approved fully populated square cache and point-phase operator. Response and covariance mathematics is preserved conditionally for future science, but no such base-v0.1 product is created. Historical implementation behavior, parameter recommendations, audits, and achieved results remain excluded.

The genuinely new Stage B work is therefore documentary and contractual: one shared scientific authority is expressed as this rationale and the companion Engineering Conformance Specification. The shared modules define notation, definitions, equations, assumptions, requirements, and falsifiable predictions. No estimator re-derivation is performed merely to make the work appear new.

2 Physical model and scientific intent

An on-the-fly observation samples the sky irregularly. The telescope and upstream processing determine which spatial information is present in those samples; gridding then estimates values on a regular destination WCS. The generic JINC motivation is spatial-frequency control: the peak-normalized Fourier partner of a circular aperture passband is a signed JINC function [3]. A practical analytic family multiplies one JINC factor by a generalized-exponential envelope and a second JINC factor. The second factor places a controlled zero while the envelope damps distant sidelobes.

The scientific object is not a nearest-neighbor average and not an ordinary positive interpolation kernel. Positive lobes, analytic zeros, and negative lobes are all part of the operator. The sign carries physical mathematical meaning: a negative lobe subtracts from both the signal numerator and the normalization, while it enters the quadratic statistic and coefficient-squared time through its square. Removing that sign would change the estimator.

The generic analytic identity is completely specified by Equations (1) and (2). The angular separation is normalized by an explicit array-associated scale s_a . The dimensionless tuple $(a_a, b_a, c_a, (r_{\max})_a)$ controls the two JINC arguments and envelope. This defines what each parameter means; it does not select TolTEC values. The Schloerb excerpt supplies a generic LMT method, not TolTEC array tuning, validation, or performance authority.

3 Scientific boundary and ownership

SCI-JINC begins with one exact PTC occurrence and the AST continuous coordinate for the same processed sample realization. PTC owns the transformed quantity, retention, cleaning realization, immutable lineage, and the one versioned registry of positive analysis/gridding coefficient families. Every family declares permission for named consumers; permission for SCI-MAP does not imply permission

for SCI-JINC. The user selects an allowed family, or an explicit versioned mode policy supplies a default. Requested, effective, observation-resolved, and realized identities remain distinct.

AST owns the coordinate, frame, WCS, parent-sample association, validity, uncertainty, and producer causes. SCI-JINC does not reconstruct pointing or infer a coordinate from row order, time tolerance, detector order, shape, or numerical equality. It consumes the AST coordinate only after the exact same-processed-sample relation is established.

SCI-JINC owns the named map-contribution admission use, local offset and radial geometry, center rounding, point phase, square membership, signed coefficient, normalization, cancellation conditioning, fixed accumulators, local map support, grouping, destination identity, and atomic publication. It applies upstream facts without changing their meanings. A producer cause has an effect only when an exact restriction names it; generic flags do not acquire a universal downstream veto.

This division follows the producer–transformer–consumer rule. ALIGN/AST, RTC, CAL, PTC, and VAL retain their own meanings. Later noise, filtering, beam, source, observing-mode, and recurrence packages retain theirs. SCI-JINC does not create those sciences at the gridding boundary.

4 Analytic kernel, point phase, and finite geometry

The AST continuous coordinate is resolved against the target WCS. SCI-JINC rounds the sample center, forms a residual phase on each axis, quantizes that phase, and selects one point-evaluated kernel matrix. Increasing `subpixel_n` refines this point-phase representation. It does not approach a pixel-area integral, because pixel-area integration is not the approved operator.

The square cache is fully populated. The parameter $(r_{\max})_a$ serves two roles: it places the first zero of the second JINC factor and fixes the square half-width. It is not a maximum evaluated radius. A square corner beyond radial $(r_{\max})_a$ remains part of the operator and receives the analytic continuation of Equation (2). A circular mask would define a different response.

The finite-map rule first tests the resolved rounded center. If that center is outside the destination domain, the occurrence contributes nowhere, even if its square would overlap the map. If the center is inside, ordinary crop removes square pixels outside the map. There is no wrap, reflection, completion, edge correction, or interior-footprint renormalization. Direct construction on a smaller map need not equal constructing a larger map and cropping it afterward.

Tie rules, bin edges, phase representatives, cache rounding, and reduction order are engineering realizations, not additional scientific branches. They must be single-valued, preserve the accepted operator, and keep their combined numerical error negligible compared with the approximately 10^{-3} relative fidelity relevant to the instrument. This contract prescribes neither bitwise reproducibility nor a universal implementation algorithm.

5 Signed estimator and cancellation

The upstream ω_i is positive. SCI-JINC supplies the signed spatial factor κ_{ip} and forms $w_{ip} = \omega_i \kappa_{ip}$. For every destination pixel it accumulates distinct N_p , C_p , and Q_p using the same occurrence membership and coefficient realization. The normalized estimate is $m_p = N_p/C_p$, not N_p/Q_p . This choice gives unit response to a constant input wherever the signed normalization is accepted.

Signed cancellation is the central validity issue. A finite negative C_p is scientifically admissible. Exact $C_p = 0$ is unavailable, not zero sky. The dimensionless statistic $\rho_p = |C_p|/D_p$ describes the degree

of cancellation without introducing a unit-bearing threshold. It is an indicator, not a universal cutoff. A very small but nonzero denominator is supportable only when the complete numerical realization demonstrates error negligible relative to the instrument-fidelity scale; otherwise the pixel is locally invalid.

The contract does not require a particular summation algorithm, contributor-count/machine-epsilon formula, reduction order, or absolute C_p/Q_p floor. Those choices belong to later engineering assessment. What remains scientific is invariant: membership, sign, equations, unit behavior, exact-cancellation rejection, finite-negative normalization, and the adequacy principle.

6 Products, grouping, and lifecycle

Every produced bundle has exactly five required roles. Three are primary accumulator planes: N_p , C_p , and Q_p . The fourth is derived m_p with its local support/validity state. The fifth is $T_p^{(\kappa^2)}$, coefficient-squared time. This time product accounts for the squared analytic coefficient divided by processed-sample frequency. It is not acquired exposure, complete temporal support, precision, validity, or significance.

Table 1: Fixed product meanings and prohibited reinterpretations.

Role	Scientific meaning	It is not
Signal numerator	Signed sum $N_p = \sum_i I_{ip} \omega_i \kappa_{ip} z_i$.	A normalized map, response, or significance plane.
Signed normalization	Signed sum $C_p = \sum_i I_{ip} \omega_i \kappa_{ip}$.	Exposure, positive weight, support, or a quantity that may be forced positive.
Quadratic accumulator	$Q_p = \sum_i I_{ip} \omega_i \kappa_{ip}^2$, required by local formal support.	Automatically variance, precision, exposure, or validity.
JINC map	$m_p = N_p/C_p$ on accepted local support, in the admitted signal unit.	Empirical significance, a source fit, or a response companion.
Coefficient-squared time	$T_p^{(\kappa^2)}$ in seconds, accounting for the squared analytic coefficient.	Physical acquired exposure, hits, white-noise-equivalent time, normalized influence, or confidence.

A whole-product failure in any required role suppresses the bundle. A local invalid pixel does not: local support is ordinary map content. The fixed schema therefore needs neither a generic role-availability object nor a placeholder product. It also creates no response, covariance, formal-weight, physical-exposure, diagnostic, or generalized provenance role.

Observation is the scientific grouping boundary. One observation may produce zero through three independent bundles, at most one for each of `a1100`, `a1400`, and `a2000`. A missing, unavailable, or unrequested array creates no placeholder and does not invalidate a different array. Contributions with different array or destination identities never merge.

Same-observation samples or processing chunks may realize one bundle only under one exact observation,

array, JINC realization, target WCS, admission/parameter/coefficient state, destination, and lifecycle generation. A chunk, worker, process, container, filename, or vector position is not a scientific product identity. Cross-observation combination is not defined. A future JINC coadd must begin from complete observation bundles under a separate scientific boundary.

7 Validity, uncertainty, and claim separation

Local formal support requires finite admitted inputs, finite N_p , C_p , and Q_p , $Q_p > 0$, $C_p \neq 0$, and numerically adequate resolved cancellation. Empirical downstream policy may narrow this state but cannot promote an invalid pixel. Outside square support, an analytic zero, and an outside rounded center are ordinary no-contribution states. Missing or ambiguous coordinate association is different: geometry is unavailable and must not be relabeled as outside support.

The formal variance and covariance equations are retained because they are part of the recovered signed-estimator science. They are conditional on the exact coefficient family and covariance assumptions. With dimensionless ω_i , Q_p/C_p^2 is dimensionless and is not signal variance. Even with inverse-square units, units alone do not prove inverse variance or independence. No formal-weight or covariance plane is therefore authorized for the fixed bundle.

Table 2: Uncertainty budget and present evidentiary state.

Source	Scientific treatment	Present state
Signal and calibration	Retain PTC/CAL quantity, calibration, response, uncertainty, quality, and lineage as producer facts.	Not re-estimated or validated by SCI-JINC.
Astrometry	Retain AST coordinate uncertainty, Jacobian availability, WCS, and causes.	No JINC astrometric performance claim.
PTC coefficient	Retain the selected family's statistic, normalization, support, uncertainty, QC, and covariance meaning.	A realized JINC-permitted family is known but not supplied by the author packet.
Kernel parameters	Bind an exact array-associated parameter-set identity.	TolTEC numerical values are known but not supplied; the route is unavailable.
Finite numerical realization	Combined error must be negligible relative to the approximately 10^{-3} instrument-fidelity scale.	Adequacy has not been tested or achieved by this draft.
Response and covariance	Preserve conditional identities for later use.	No base-v0.1 product and no achieved response/covariance claim.
Observational performance	Requires later authorized numerical and observational validation.	Not performed.

Table 3: Claim layers remain separate.

Layer	What it would establish	Status of this draft
Algebraic contract correctness	Internal consistency of identities, equations, units, limits, and predictions.	Drafted and mechanically reviewable; not yet owner-approved or frozen.
Implementation conformity	A candidate realizes the binding requirements.	Not assessed; no candidate inspected.
Representation fidelity	Stored or transmitted products preserve the scientific identities and meanings.	Not assessed.
Numerical and observational validation	Realized error, response, covariance, photometry, and observation behavior satisfy authorized profiles.	Not performed.
Achieved performance	Measured scientific performance under named conditions.	No claim.
Readiness and production authorization	Operational qualification and explicit owner authorization.	No claim.

8 Limiting cases and interpretive safeguards

The signed estimator has simple limiting cases that expose its meaning without requiring an implementation-specific realization. For a constant field, $z_i = z_0$ factors out of the numerator, so $N_p = z_0 C_p$ and $m_p = z_0$ wherever the normalization is accepted. For one contributor with nonzero κ_{ip} , the signed coefficient cancels between numerator and normalization and the estimate returns that contributor's signal. These are normalization identities, not achieved photometric-validation claims.

Common positive scaling of every ω_i changes the three primary accumulators together but leaves m_p , ρ_p , and the normalized operator unchanged. This distinguishes coefficient normalization from scientific signal normalization. A signal-unit change has its own transformation: when an exact family truly represents inverse variance, scaling the signal by λ requires scaling the coefficient by λ^{-2} . The map then scales by λ and the conditional formal weight by λ^{-2} .

Table 4: Interpretation of representative limiting cases.

Case	Algebraic result	Scientific safeguard
Constant input	$N_p = z_0 C_p$ and $m_p = z_0$ on accepted support.	Tests the signed N/C normalization, not achieved calibration.
One nonzero contributor	$m_p = z_i$ for either sign of κ_{ip} .	A negative lobe is not an invalid sample.
Analytic zero	The occurrence changes none of N_p , C_p , Q_p , or $T_p^{(\kappa^2)}$.	The zero remains distinguishable from outside support.

Case	Algebraic result	Scientific safeguard
Finite negative lobe	N_p and C_p retain sign; Q_p and time receive positive squared terms.	Absolute value or clipping changes the operator.
Finite negative normalization	$m_p = N_p/C_p$ remains algebraically available if every other gate passes.	Denominator sign alone is not a validity cut.
Exact cancellation	$C_p = 0$ makes m_p unavailable.	Never substitute zero sky, infinity, or a valid non-finite value.
Near cancellation	Availability depends on the demonstrated total numerical error relative to the 10^{-3} scale.	No universal ρ_p threshold or unit-bearing floor is inferred.
Outside rounded center	All $I_{ip} = 0$, even with footprint overlap.	The center-domain gate precedes square support.
In-map cropped edge	Retained pixels use their actual membership and signed N/C normalization.	No wrap, completion, edge correction, or JINC-then-crop requirement.
Common coefficient scale	N_p , C_p , and Q_p scale together; m_p and ρ_p do not.	Coefficient normalization cannot silently change the map.

Coefficient-squared time has a different limiting structure. Its term is zero at an analytic zero and positive for either sign of a nonzero JINC lobe; changing ω_i alone does not change it. Those facts are exactly why the product cannot be reinterpreted as weighted exposure, precision, or normalized influence.

The edge rules deliberately make finite-map construction part of the observation estimator. An occurrence with an outside rounded center is excluded before footprint evaluation, while an occurrence with an inside center uses a cropped square. Therefore a larger-map JINC construction followed by cropping is not required to equal direct construction on a smaller map. This is a scientific boundary decision, not a numerical defect.

9 Future evidence route without a validation claim

The companion engineering view converts the shared requirements and predictions into an evidence checklist, but no evidence is evaluated during Stage B authorship. A future assessment must first bind the exact coefficient family, TolTEC parameter set, observation, array, WCS, plan, destination, and lifecycle generation. Without those identities, a numerical example cannot stand in for the scientific route.

Algebraic checks should begin with hand-constructed constant, one-contributor, two-contributor, analytic-zero, negative-lobe, and cancellation cases. Geometry checks should then exercise every map edge and corner, half-pixel tie, phase-bin edge, `subpixel_n=1`, square-cache corner, and center immediately inside and outside the finite domain. These constructions establish the operator being tested; they do not prescribe one adequate tie or bin algorithm.

Numerical adequacy requires a disclosed combined error budget. Analytic function evaluation, phase

quantization, cache/index realization, and accumulation/reduction order must be considered together. Agreement between two executions is useful evidence but is not itself the scientific criterion: two paths could agree while sharing the same wrong membership or normalization. Conversely, harmless non-bitwise differences may conform when the combined error remains negligible relative to the instrument scale.

Table 5: Future evidence layers and their present limits.

Layer	Representative evidence	What it cannot establish alone
Authority binding	Exact parameter, coefficient, observation, WCS, plan, destination, and lifecycle identities.	Numerical correctness or achieved performance.
Algebra and units	Hand cases, scaling identities, zero limit, signed lobes, and conditional unit classifications.	Implementation conformity without a bound candidate.
Geometry and membership	Tie/bin neighborhoods, square corners, outside-center gate, inside-edge crop, and exact AST association cases.	Observational response or photometry.
Numerical adequacy	Disclosed error budget and adversarial cancellation cases against the approximately 10^{-3} scale.	Bitwise reproducibility or readiness.
Bundle/lifecycle	Array permutation, missing-array, destination collision, chunking, atomic failure, and forbidden-role cases.	Representation fidelity without stored-product evidence.
Representation	Exact stored identities, units, WCS, support state, and role semantics.	Observational validation or production authorization.
Observation	Separately authorized response, covariance, photometric, and science-performance evidence.	Readiness or production authority without an explicit decision.

A fail-closed test may demonstrate that missing parameter or coefficient authority prevents a bundle. It may not manufacture the missing authority in order to exercise a numerical path. Likewise, a future response or covariance study must first receive separate product authority; base-v0.1 conditional equations do not turn an observed output into an authorized companion product.

10 Typed unavailability and next scientific dependencies

The contract is scientifically complete at the generic level, but two inputs needed for any numerical route are known and not supplied. First, an exact PTC coefficient family must be registered, selected, realized, and explicitly permitted for SCI-JINC. Second, an exact TolTEC array-associated parameter set must be scientifically authorized. Neither absence is repaired by memory or software behavior.

No inherited 45 m denominator, inherited shape values, or mode-dependent $r_{\max}$ values are authorized. No value in the generic Schloerb excerpt is a TolTEC default, recommendation, optimum, validation result, or performance claim. Selecting TolTEC parameters is a separate scientific exercise with a named objective and appropriate TolTEC/LMT beam, response, and/or noise evidence.

No unresolved numbered scientific-scope question remains in the approved packet, and this draft prepared without implementation access finds no internal contradiction that requires a new owner question. Response/covariance companions, physical exposure, cross-observation coaddition, and broader diagnostics or provenance remain deferred until a concrete scientific use separately authorizes them.

A Canonical notation and units

The symbols in Table 6 are normative in both views of this contract. The generic signal unit is denoted by U . The admitted upstream boundary presently names a nonpolarimetric total-intensity-equivalent quantity in mJy/beam ; the symbol U keeps the algebra explicit and does not turn a component token into a physical Stokes assertion.

Table 6: Canonical notation.

Symbol	Meaning and domain	Unit
a	Stable TolTEC array identity, exactly one of a1100 , a1400 , or a2000 .	identity
i	Exact admitted PTC occurrence on stable RTC output sample n , with observation, detector occurrence/UID, PTC application generation, segment, array, and immutable parents.	identity
p	Target pixel in one exact JINC destination WCS.	identity
z_i	Exact finite transformed signal Z_i^{PTC} .	U
ω_i	Positive, finite, PTC-produced analysis/gridding coefficient from one exact registered family/version that permits SCI-JINC.	1 or U^{-2} , as declared by its family
κ_{ip}	Signed, dimensionless analytic JINC coefficient at the selected point phase. Finite positive, exact-zero, and finite negative values are distinct legitimate analytic results.	1
w_{ip}	Complete signed contribution coefficient, $w_{ip} = \omega_i \kappa_{ip}$.	unit of ω_i
I_{ip}	Boolean membership after sample admission, exact AST association, rounded-center domain admission, local square geometry, and all pixel-local gates.	1
s_a	Explicit angular radial scale associated with stable array a ; finite and strictly positive in any numerical realization.	angle
r	Angular separation between the realized sample phase and destination pixel evaluation point in the target WCS.	angle
r'_a	Dimensionless radius r/s_a , finite and nonnegative.	1
$a_a, b_a, c_a, (r_{\max})_a$	Ordered, dimensionless analytic shape tuple for array a ; every member is finite and strictly positive in an authorized numerical realization.	1
θ_a	Complete kernel parameter identity $(s_a, a_a, b_a, c_a, (r_{\max})_a)$, including owner, source, version, units, array association, and lifecycle state.	mixed, as above
N_p	Signed signal numerator.	$U[\omega]$
C_p	Signed normalization; it may be finite and negative, but exact zero is never accepted.	$[\omega]$

Symbol	Meaning and domain	Unit
Q_p	Quadratic accumulator; finite and strictly positive on accepted local support.	$[\omega]$
m_p	Normalized estimate N_p/C_p on accepted local support.	U
D_p	Absolute-term sum $\sum_i I_{ip} \omega_i\kappa_{ip} $, used only as construction state for conditioning.	$[\omega]$
ρ_p	Dimensionless cancellation statistic $ C_p /D_p$.	1
A_{pi}	Fixed-state normalized linear operator row $I_{ip}\omega_i\kappa_{ip}/C_p$.	1
$f_{s,i}$	Effective processed-sample frequency, finite and strictly positive.	s^{-1}
$T_p^{(\kappa^2)}$	Required <code>jinc_coefficient_squared_time</code> accumulator.	s

In-memory indices are zero-based. Persisted FITS/WCS pixel coordinates are one-based. Memory order never supplies axis sign, handedness, orientation, or wrapping.

B Normative definitions and boundary

SCI-JINC observation estimator.

A signed spatial-kernel transformer that consumes exact PTC signal occurrences and their exact AST coordinates and forms at most one complete JINC bundle for each admitted and requested stable array in one observation. It is a sibling alternative to ordinary SCI-MAP, not a downstream stage of SCI-MAP.

Exact occurrence.

The scientific identity that jointly fixes observation, detector occurrence and UID, stable RTC output sample, PTC product and application generation, PTC segment, stable array/group, time attribute, and immutable parent chain. Row position, nearest time, tolerance, shape, detector order, or numerical equality is not a substitute.

Exact AST association.

The relation binding z_i to the AST-owned `SCI-AST:rtc_output_grid_coordinates@1` continuous coordinate for the same processed sample realization in the exact target JINC WCS. Its data-model join is an engineering choice; its scientific identity is not.

JINC sample admission.

The decision of the immutable `SCI-JINC:jinc_map_contribution@1` profile that one exact upstream occurrence may be considered by JINC. It neither implies rounded-center domain admission nor pixel support.

Rounded-center domain admission.

The JINC-owned occurrence-level gate applied after center rounding and before any footprint evaluation. The resolved rounded center must lie within the finite destination pixel domain. An outside center makes $I_{ip} = 0$ for every p , even when the square footprint would overlap the map.

Point phase.

The single-valued residual between the exact continuous sample coordinate and the resolved rounded center, quantized on each axis to select one point-evaluated kernel matrix. It is not a pixel-area average.

Square support.

The fully populated square cache whose half-width is fixed by $(r_{\max})_a$. Every resolved square pixel is evaluated, including corners with radius larger than $(r_{\max})_a$; there is no circular membership predicate.

Sample-pixel membership.

The Boolean proposition $I_{ip} = 1$ after sample admission, exact coordinate association, rounded-center domain admission, square placement, finite local geometry, and every pixel-local precondition. A contract-defined analytic zero is a member with zero contribution; outside support is nonmembership; a finite negative coefficient is an ordinary signed member.

Local formal support.

The state in which the required inputs and accumulators are finite, $Q_p > 0$, $C_p \neq 0$, and cancellation is resolved with total numerical error negligible relative to the approximately 10^{-3} instrument-fidelity scale. A named empirical policy may narrow this state but never promote it.

Complete observation bundle.

The atomic fixed-role product for one observation, stable array, exact JINC plan and realization, target WCS, and lifecycle generation. It contains exactly N_p , C_p , Q_p , derived m_p with local support/validity, and $T_p^{(\kappa^2)}$.

Whole-bundle failure.

Failure to form or publish any required whole-product role. It suppresses the affected bundle without a partial role, placeholder, synthetic empty-array product, or completion override. Pixel-local invalid support is ordinary bundle content and is not whole-bundle failure.

Typed numerical unavailability.

The fail-closed state in which a numerical JINC route cannot be realized because an exact authorized TolTEC kernel parameter set or an exact registered, selected, realized, JINC-permitted PTC coefficient family/payload is absent. It is not repaired by a hidden default, unity coefficient, alternate family, ordinary MAP permission, inherited scale, or implementation value.

Scientific destination identity.

The product identity formed from observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Worker, thread, process, container, filename, vector position, and network order are not scientific identity.

C Canonical equations

C.1 Peak-normalized analytic family

Let J_1 be the first-order Bessel function of the first kind. Define the collision-free peak-normalized function

$$\mathcal{J}(x) = \begin{cases} 2J_1(x)/x, & x \neq 0, \\ 1, & x = 0. \end{cases} \quad (1)$$

For $r'_a = r/s_a$ and $j_{1,1} = 3.831706$, the generic SCI-JINC coefficient is

$$\kappa_a(r'_a; a_a, b_a, c_a, (r_{\max})_a) = \mathcal{J}\left(\frac{2\pi r'_a}{a_a}\right) \exp\left[-\left(\frac{2r'_a}{b_a}\right)^{c_a}\right] \mathcal{J}\left(\frac{j_{1,1}r'_a}{(r_{\max})_a}\right). \quad (2)$$

Equation (2) has $\kappa_a(0) = 1$. a_a scales the first JINC argument, b_a scales the generalized-exponential envelope, c_a is its exponent, and $(r_{\max})_a$ places the first zero of the second JINC factor. Analytic zeros and finite signed lobes are retained. The square-cache rule, rather than a radial cutoff, determines membership.

C.2 Signed estimator

For the exact occurrence population of one bundle,

$$w_{ip} = \omega_i \kappa_{ip}, \quad (3)$$

$$N_p = \sum_i I_{ip} \omega_i \kappa_{ip} z_i, \quad (4)$$

$$C_p = \sum_i I_{ip} \omega_i \kappa_{ip}, \quad (5)$$

$$Q_p = \sum_i I_{ip} \omega_i \kappa_{ip}^2, \quad (6)$$

$$m_p = \frac{N_p}{C_p}, \quad (7)$$

$$A_{pi} = \frac{I_{ip} \omega_i \kappa_{ip}}{C_p}. \quad (8)$$

All coupled accumulator terms for one contribution use the same I_{ip} and the same realized κ_{ip} identity. The normalization is signed. A finite negative C_p is admissible; exact $C_p = 0$ is not a zero-sky result.

C.3 Conditioning and local support

Define construction-state quantities

$$D_p = \sum_i I_{ip} |\omega_i \kappa_{ip}|, \quad (9)$$

$$\rho_p = \frac{|C_p|}{D_p}, \quad D_p > 0. \quad (10)$$

For exact arithmetic, $0 \leq \rho_p \leq 1$. Accepted local support requires finite contributors, finite N_p , C_p , and Q_p , $Q_p > 0$, $C_p \neq 0$, and a demonstration that total numerical error is negligible compared with the approximately 10^{-3} relative fidelity relevant to the instrument. Neither D_p nor an error estimate is a persistent base-v0.1 product.

C.4 Coefficient-squared time

The sole authorized base-v0.1 time-support product is

$$T_p^{(\kappa^2)} = \sum_i I_{ip} \frac{\kappa_{ip}^2}{f_{s,i}}. \quad (11)$$

It is nonnegative and has unit seconds. The squared object is κ_{ip} , not ω_i or w_{ip} .

C.5 Conditional mathematics retained for future science

The following identities do not create base-v0.1 products. If and only if an exact PTC family establishes $\omega_i = \text{Var}(z_i)^{-1}$ for mutually independent admitted occurrences,

$$\text{Var}_{\text{formal}}(m_p) = \frac{Q_p}{C_p^2}, \quad (12)$$

$$W_{\text{formal},p} = \frac{C_p^2}{Q_p}, \quad (13)$$

$$\text{Cov}(m_p, m_{p'}) = \frac{\sum_i I_{ip} I_{ip'} \omega_i \kappa_{ip} \kappa_{ip'}}{C_p C_{p'}}. \quad (14)$$

For a general available upstream covariance,

$$\mathbf{C}_{\text{JINC}} = \mathbf{A}_{\text{JINC}} \mathbf{C}_{\text{PTC}} \mathbf{A}_{\text{JINC}}^T. \quad (15)$$

Inverse-square units alone do not establish inverse variance, independence, precision, covariance authority, or significance. Response, variance, formal-weight, and covariance companions remain deferred until a concrete scientific use separately authorizes them.

D Assumptions, validity classes, and explicit unavailability

Table 7: Normative assumptions and boundary states.

ID	Statement
SCI-JINC-ASM-001	The contract concerns one observation at a time and a nonpolarimetric total-intensity-equivalent transformed signal. It does not authorize a physical Stokes interpretation from a component token.
SCI-JINC-ASM-002	PTC and AST meanings, identities, validity facts, causes, uncertainty, and immutable ancestry are producer-owned facts. JINC uses them without reconstructing or redefining them.
SCI-JINC-ASM-003	An admitted ω_i is finite and strictly positive and comes from one exact PTC family/version that explicitly permits SCI-JINC. No registered family or realized payload is supplied by the scientific author packet.
SCI-JINC-ASM-004	Equations (12)–(15) are conditional identities only. Their assumptions are not inferred from units, names, or another consumer’s permission.
SCI-JINC-ASM-005	The analytic family and parameter meanings are approved, but no TolTEC numerical realization of s_a , a_a , b_a , c_a , $(r_{\max})_a$, pixel size, or subpixel_n is scientifically supplied or recommended by this contract.
SCI-JINC-ASM-006	Point-phase evaluation and fully populated square support are the scientific operator. Pixel-area integration, a circular cutoff, footprint-overlap admission, and JINC-then-crop equivalence are not alternative branches.

ID	Statement
SCI-JINC-ASM-007	Total numerical error includes finite arithmetic, accumulation/reduction order, analytic-function evaluation, phase quantization, and cache/index realization. Adequacy is judged against the approximately 10^{-3} relative instrument-fidelity scale; no stronger universal precision claim follows.
SCI-JINC-ASM-008	Same-observation chunking may be an engineering realization of one bundle, but a chunk is not a scientific product identity. Cross-observation combination is outside base v0.1.
SCI-JINC-ASM-009	This Stage B draft has inspected no implementation candidate, configuration, product, test, audit, validation, reduction, or production evidence.

Every presently missing fact is classified exactly once in Table 8.

Table 8: Missing-fact classification.

Class	Fact	Consequence
Approved	Generic analytic identity, signed estimator, point-phase square operator, center-domain gate, conditioning principle, fixed bundle, grouping, and ownership boundary.	These are normative in v0.1.
Known but not supplied	A registered, selected, and realized JINC-permitted PTC coefficient family and payload.	No numerical bundle is formed without it; no unity or alternate-family fallback.
Known but not supplied	A scientifically authorized TolTEC array-associated numerical parameter set.	No numerical bundle is formed; no inherited scale, shape, extent, or mode-dependent value is used.
Unresolved decision	None within the approved base-v0.1 scientific scope.	No owner question is raised by this draft.
Implementation unassessed	Representation, interfaces, algorithms, storage, parallel realization, and implementation behavior.	No conformity or representation-fidelity claim.
Validation not performed	Algebraic execution, numerical adequacy, observational response, covariance, photometry, performance, readiness, and production operation.	No achieved result, readiness, or production claim.

Response, covariance, formal weight, physical exposure, cross-observation coaddition, standalone support or availability, diagnostics, and generalized provenance are deferred or outside base v0.1, rather than silently missing members of the fixed bundle.

E Binding scientific requirements

The key words *shall*, *shall not*, *required*, and *unavailable* are normative. These requirements define scientific conformance; they do not assert that any implementation satisfies them.

SCI-JINC-REQ-001. SCI-JINC shall define a signed spatial-kernel estimator for one observation and stable array. It shall consume exact PTC and AST parents directly, consume no SCI-MAP product, and inherit no ordinary SCI-MAP normalization, support, exposure, validity, response, covariance, product, or coaddition rule by analogy.

SCI-JINC-REQ-002. Every candidate contribution shall bind one exact occurrence and the AST continuous coordinate associated with the same processed sample realization in the exact target JINC WCS. Row order, detector order, nearest time, tolerance, shape, cardinality, or numerical coordinate equality shall not establish that association.

SCI-JINC-REQ-003. The transformed signal z_i shall retain its exact PTC quantity, unit, availability, application generation, retention state, immutable parents, and producer causes. SCI-JINC shall spatially transform that quantity without creating calibration, conditioning, astrometry, empirical noise, or source interpretation.

SCI-JINC-REQ-004. Every ω_i shall come from one exact family/version in the single PTC-owned positive analysis/gridding coefficient registry. The family shall explicitly permit SCI-JINC, and its requested, effective, observation-resolved, and realized identities, payload, QC, normalization, support, provenance, covariance meaning, uncertainty, and prohibited meanings shall remain bound.

SCI-JINC-REQ-005. A numerical route shall fail closed when coefficient selection is missing without an authorized versioned mode default, the family is unregistered, JINC permission is absent, or the payload/QC/identity is absent, unavailable, incompatible, duplicate, or ambiguous. SCI-JINC shall not infer unity, another family, ordinary MAP permission, **sens**, loading, scatter, inverse variance, precision, or significance.

SCI-JINC-REQ-006. After the producer-owned coefficient decision, SCI-JINC shall admit only a finite strictly positive ω_i . Zero, negative, non-finite, or unrepresentable upstream coefficient values shall not be deposited and shall retain the exact profile-defined disposition and cause.

SCI-JINC-REQ-007. Every numerical kernel route shall bind one exact scientifically authorized array-associated parameter-set identity containing finite positive s_a , a_a , b_a , c_a , $(r_{\max})_a$, finite positive pixel size, and effective integer **subpixel_n** ≥ 1 . Missing authority shall make the affected route unavailable without fallback.

SCI-JINC-REQ-008. The peak-normalized analytic primitive shall be $\mathcal{J}(x) = 2J_1(x)/x$ for $x \neq 0$ with the removable limit $\mathcal{J}(0) = 1$, as in Equation (1).

SCI-JINC-REQ-009. The analytic coefficient shall follow Equation (2), including its fixed parameter ordering and $r'_a = r/s_a$. It shall have peak one, preserve analytic zeros, and preserve every finite positive and negative lobe of the complete product.

SCI-JINC-REQ-010. The sample center shall be resolved by a single-valued rounding rule, its two-axis residual phase shall be quantized, and one point-evaluated kernel matrix shall be selected. This operator shall not be represented as or replaced by pixel-area integration.

SCI-JINC-REQ-011. Tie handling, phase intervals, bin edges, representatives, cache-index rounding, analytic-function evaluation, and accumulation order may be engineering choices only when they are single-valued, preserve the accepted point-phase operator, and meet the numerical-adequacy requirement.

SCI-JINC-REQ-012. $(r_{\max})_a$ shall place the first zero of the second JINC factor and fix the fully populated square-cache half-width. Every resolved square pixel, including a corner with radius greater than $(r_{\max})_a$, shall be evaluated; no circular cutoff or radial membership predicate shall be applied.

SCI-JINC-REQ-013. Before any footprint evaluation, the resolved rounded cache center shall lie within the finite destination pixel domain. An outside center shall set $I_{ip} = 0$ for every pixel, even when its square would overlap the map; footprint-overlap admission is prohibited.

SCI-JINC-REQ-014. For an admitted in-map center, square pixels outside the finite map shall be absent. The route shall not wrap, reflect, complete the footprint, substitute an interior normalizer, renormalize the edge, or require JINC-then-crop equivalence.

SCI-JINC-REQ-015. Every evaluated κ_{ip} shall be finite. A finite negative coefficient is normal and shall retain its sign; an analytic zero is a contract-defined zero contribution distinct from outside support; a non-finite coefficient shall fail the affected required bundle before deposition rather than be clipped, skipped, or zero-filled.

SCI-JINC-REQ-016. All coupled accumulator terms for one contribution shall use the same exact occurrence, AST association, sample-admission decision, rounded-center decision, sample-pixel membership, phase, and κ_{ip} identity.

SCI-JINC-REQ-017. The required accumulators shall be distinct and shall obey Equations (4)–(6). Negative lobes shall keep their sign in N_p and C_p and enter Q_p quadratically.

SCI-JINC-REQ-018. The published signal estimate shall be $m_p = N_p/C_p$ and the fixed-state normalized operator shall be Equation (8). No Q_p denominator or other local measurement model shall be substituted.

SCI-JINC-REQ-019. Local formal support shall require finite admitted inputs, finite N_p , C_p , and Q_p , $Q_p > 0$, $C_p \neq 0$, and numerically adequate resolved cancellation. A rejected pixel shall carry local invalid support in the `jinc_map` role and shall not receive a finite substitute.

SCI-JINC-REQ-020. A finite negative C_p shall be admissible. Exact $C_p = 0$ shall be locally invalid and unavailable, never interpreted as zero sky, infinity, or a valid non-finite map value.

SCI-JINC-REQ-021. SCI-JINC shall retain the dimensionless conditioning statistic ρ_p of Equation (10). A finite nonzero result shall be supported only when total numerical error is demonstrably negligible relative to the approximately 10^{-3} instrument-fidelity scale; otherwise near cancellation shall be locally invalid.

SCI-JINC-REQ-022. No universal ρ_p cutoff, unit-bearing C_p or Q_p floor, contributor-count/machine-epsilon formula, prescribed summation algorithm, fixed reduction order, bitwise reproducibility, or stronger precision shall be treated as scientific authority.

SCI-JINC-REQ-023. Total numerical error from finite arithmetic, accumulation and reduction order, analytic-function evaluation, phase quantization, and cache/index realization shall remain negligible

relative to the approximately 10^{-3} instrument-fidelity scale while preserving exact membership and the point-phase square operator.

SCI-JINC-REQ-024. The unit relations in Table 6 and the two allowed ω_i unit classes shall be preserved. Common positive scaling of all ω_i shall leave m_p , ρ_p , and A_{pi} invariant; signal-unit rescaling shall transform the estimator consistently.

SCI-JINC-REQ-025. Equations (12)– (15) may receive a formal statistical interpretation only when the exact PTC family and covariance assumptions establish it. Inverse-square units or historical labels alone shall not establish variance, precision, covariance, or significance.

SCI-JINC-REQ-026. The required time-support product shall be exactly Equation (11). It shall square κ_{ip} , not ω_i or w_{ip} , and shall have unit seconds.

SCI-JINC-REQ-027. `jinc_coefficient_squared_time` shall be described only as method-specific coefficient-squared temporal accounting. It shall not be interpreted as physical acquired exposure, valid-original exposure, complete temporal support, hits, normalized influence, white-noise-equivalent time, precision, validity, confidence, or significance.

SCI-JINC-REQ-028. Every produced base-v0.1 bundle shall contain exactly five required roles: `jinc_signal_numerator`, `jinc_signed_normalization`, `jinc_quadratic_accumulator`, derived `jinc_map` with local support/validity, and `jinc_coefficient_squared_time`.

SCI-JINC-REQ-029. Failure to form or publish any required whole-product role shall suppress the affected bundle atomically. Pixel-local zero, insufficient, cancelled, or otherwise invalid support shall remain ordinary bundle content and shall not suppress an otherwise complete role or create a role-availability object.

SCI-JINC-REQ-030. One observation may produce zero through three independent bundles, with at most one for each stable array admitted and requested under the exact JINC realization. A missing, unavailable, or unrequested array shall create no placeholder or empty-array product and shall not invalidate another array’s bundle.

SCI-JINC-REQ-031. Each bundle shall be independently identified by observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Contributions with different array or destination identities shall never merge.

SCI-JINC-REQ-032. The complete destination identity shall be resolved before allocation, numerical mutation, output side effects, or parallel execution. Ambiguous, duplicate, or aliased destination ownership shall fail the affected bundle without selecting a winner or publishing a partial result.

SCI-JINC-REQ-033. Same-observation samples or chunks may accumulate incrementally only when observation, stable array, JINC plan and realization, target WCS, admission/parameter/coefficient state, destination, and lifecycle generation are identical. Chunk, worker, process, container, filename, and vector position shall not create product identity.

SCI-JINC-REQ-034. Base v0.1 shall authorize no cross-observation combination. Any future JINC coadd shall require a separately authorized boundary over complete observation bundles and shall not infer accumulator addition, normalized-map combination, or ordinary SCI-MAP coaddition.

SCI-JINC-REQ-035. The immutable `SCI-JINC:jinc_map_contribution@1` evaluation shall remain distinct from PTC signal availability, PTC retention, coefficient availability/QC, AST coordinate validity

and association, finite signal, finite positive ω_i , parameter availability, rounded-center admission, pixel support, finite κ_{ip} , cancellation, and final bundle validity.

SCI-JINC-REQ-036. Only a fact named by an exact restriction shall control JINC admission. Producer causes and transitive influence shall be preserved, but a generic flag or cause shall have no universal veto. Outside support, an analytic zero, or an outside rounded center shall be ordinary no-contribution; a finite negative lobe shall not create a cause.

SCI-JINC-REQ-037. ALIGN/AST, RTC, CAL, PTC, and VAL shall retain the meanings they own. SCI-JINC shall own only its admission use, local geometry, point phase, square support, signed coefficient, normalization, conditioning, fixed accumulators, local support, grouping, destination, and atomic bundle publication.

SCI-JINC-REQ-038. Base v0.1 shall publish no response, variance, formal-weight, covariance, empirical-noise, or significance product. The conditional mathematics in Section C.5 shall remain preserved for a future concrete scientific use without being promoted to a present role.

SCI-JINC-REQ-039. Base v0.1 shall publish no physical-exposure product. Any future exposure role shall require separate authority for exact original- occurrence lineage and semantics and shall not reinterpret $T_p^{(\kappa^2)}$.

SCI-JINC-REQ-040. Base v0.1 shall create no standalone support or availability role, generic optional or conditionally required role mechanism, detailed operational-cause archive, diagnostic role, generalized provenance product, or per-pixel or per contribution provenance system.

SCI-JINC-REQ-041. No TolTEC numerical parameter realization, recommendation, optimum, hidden default, or fallback is authorized by v0.1. In particular, an inherited 45 m denominator, inherited shape values, or mode-dependent $r_{\max}$ values shall not be treated as scientific authority.

SCI-JINC-REQ-042. A future conformance assessment shall exercise every prediction in Section F, disclose its numerical acceptance policy, and separate algebraic correctness from implementation conformity, representation/response fidelity, numerical and observational validation, achieved performance, readiness, and production authorization. This Stage B draft establishes none of those later claims.

F Falsifiable predictions and edge cases

These predictions are normative tests for a future authorized conformance program. Listing them is not evidence that they have been executed or passed.

Table 9: Stable SCI-JINC predictions.

Prediction	Falsifiable expected result
SCI-JINC-PRED-001	A constant finite input $z_i = z_0$ returns $m_p = z_0$ on every accepted local support, including a cropped edge pixel.
SCI-JINC-PRED-002	One admitted contributor with nonzero finite κ_{ip} returns $m_p = z_i$, whether the coefficient is positive or negative.

Prediction	Falsifiable expected result
SCI-JINC-PRED-003	A hand-computed two-contributor case with unequal z_i , ω_i , and signed κ_{ip} matches the separate N_p , C_p , Q_p , m_p , and $T_p^{(\kappa^2)}$ equations.
SCI-JINC-PRED-004	Multiplying every ω_i by the same positive factor scales N_p , C_p , and Q_p together and leaves m_p , ρ_p , and A_{pi} unchanged.
SCI-JINC-PRED-005	Under a true inverse-variance family, rescaling $z_i \mapsto \lambda z_i$ and $\omega_i \mapsto \omega_i/\lambda^2$ produces $m_p \mapsto \lambda m_p$ and conditional formal weight $W_{\text{formal},p} \mapsto W_{\text{formal},p}/\lambda^2$.
SCI-JINC-PRED-006	The analytic evaluator returns $\kappa_a(0) = 1$ through the removable limit, not a non-finite value or an implementation-dependent substitute.
SCI-JINC-PRED-007	An analytic $\kappa_{ip} = 0$ changes none of N_p , C_p , Q_p , or $T_p^{(\kappa^2)}$ while remaining distinguishable from outside square support.
SCI-JINC-PRED-008	A finite negative lobe retains its sign in N_p and C_p , enters Q_p and $T_p^{(\kappa^2)}$ quadratically, and is not clipped, absolutized, omitted, or flagged solely for being negative.
SCI-JINC-PRED-009	Exact cancellation $C_p = 0$ yields local invalid support and no finite substitute, even when N_p and Q_p are finite.
SCI-JINC-PRED-010	A finite negative C_p with otherwise adequate support yields the signed normalized estimate and is not rejected merely for the denominator sign.
SCI-JINC-PRED-011	A near-cancellation construction whose numerical error is demonstrated negligible relative to the 10^{-3} scale remains eligible; no universal ρ_p cutoff rejects it.
SCI-JINC-PRED-012	A near-cancellation construction lacking that adequacy demonstration is locally invalid, even when finite arithmetic happens to return a nonzero C_p .
SCI-JINC-PRED-013	A non-finite required signal, geometry, coefficient evaluation, accumulator, or sample frequency cannot be masked by multiplication by zero or silently skipped; the exact local or whole-bundle failure rule applies.
SCI-JINC-PRED-014	Square-cache corners whose radius exceeds $(r_{\max})_a$ are evaluated by Equation (2) and may contribute.
SCI-JINC-PRED-015	Two points on opposite sides of a hypothetical circular $r_{\max}$ boundary but inside the same resolved square are both governed by square membership; no radial cutoff appears.
SCI-JINC-PRED-016	Effective <code>subpixel_n=1</code> selects one single-valued point-phase matrix and does not become a pixel-area average.
SCI-JINC-PRED-017	Coordinates immediately around every half-pixel tie and phase-bin edge select a single-valued realization whose total error meets the numerical-adequacy requirement; no bitwise cross-realization identity is presumed.
SCI-JINC-PRED-018	A rounded center immediately outside each map edge or corner changes none of the four accumulators for any pixel, even when its square footprint overlaps the map.

Prediction	Falsifiable expected result
SCI-JINC-PRED-019	A rounded center immediately inside an edge uses only retained square pixels with no wrap, reflection, completion, renormalization, or edge correction.
SCI-JINC-PRED-020	A supported constant field at an in-map cropped edge still returns the constant through the actual retained N_p/C_p terms.
SCI-JINC-PRED-021	$T_p^{(\kappa^2)}$ is unchanged by changing ω_i alone, receives zero from an analytic zero, and receives a positive term from either sign of nonzero κ_{ip} .
SCI-JINC-PRED-022	Missing coefficient selection without an authorized default, an unregistered family, missing JINC permission, or an unavailable/mismatched payload prevents the affected numerical bundle without unity or alternate-family fallback.
SCI-JINC-PRED-023	Absence of an authorized TolTEC parameter set prevents the numerical bundle; no inherited 45 m denominator, shape values, mode-dependent $r_{\max}$, or other implementation value is substituted.
SCI-JINC-PRED-024	Permuting arrays, networks, detector rows, map groups, or container positions does not change array-to-parameter or array-to-bundle identity; a missing array creates no placeholder and does not invalidate another array.
SCI-JINC-PRED-025	Ambiguous, duplicate, or aliased destination ownership fails the affected bundle before mutation and produces no partial winner.
SCI-JINC-PRED-026	Valid decompositions of the same observation into chunks select the same scientific memberships and equations and agree within the disclosed numerical-adequacy policy; bitwise equality is not required.
SCI-JINC-PRED-027	An attempted cross-observation accumulator or map combination is rejected as outside base v0.1 rather than being assigned an inferred coadd rule.
SCI-JINC-PRED-028	A missing, duplicate, or ambiguous exact signal-coordinate association prevents the affected contribution and is not reclassified as ordinary outside support.
SCI-JINC-PRED-029	Failure to form any one of the five required whole-product roles suppresses the bundle, whereas a locally invalid pixel does not suppress an otherwise complete role.
SCI-JINC-PRED-030	The base bundle contains no response, covariance, formal-weight, physical-exposure, standalone availability/support, diagnostic, generalized provenance, or placeholder product.
SCI-JINC-PRED-031	With dimensionless ω_i , Q_p/C_p^2 remains dimensionless and is not reported as signal variance or used to create a formal-weight product.
SCI-JINC-PRED-032	The JINC route consumes no SCI-MAP product and no ordinary MAP admission, validity, exposure, support, response, covariance, or coaddition decision is substituted for a JINC-owned decision.

F.1 Prediction-to-requirement trace

Predictions	Governing requirements
001–003	016–020, 026
004–005	024–025
006–008	008–009, 015, 017, 026
009–012	019–023
013	003, 006–007, 015, 019, 026, 029
014–017	010–012, 023
018–020	013–014, 016–020
021	026–027
022–023	004–007, 041
024–027	030–034
028	002, 035–036
029–030	028–029, 038–040
031	024–025, 038
032	001, 035–038

References

- [1] SCI-JINC scientific owner and Stage A investigators, *Signed-Coefficient JINC Observation Map-maker Scope Brief*, v0.1, 2026-08-28, exact object admitted by the SCI-JINC author packet.
- [2] Independent scientific-contract audit, *Independent Core: JINC Deposition, Normalization, Response, and Product Contract*, frozen 2026-08-03, used only under the SCI-JINC supersession cover.
- [3] F. Peter Schloerb, *Spectral Line Data Reduction at the Large Millimeter Telescope*, version 1.1, 2019-07-23, method excerpt from source pages 15–19, used only under the SCI-JINC LMT method reference cover.